Humpback Whale Individual Identification

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Abstract

This project addresses the challenge of identifying individual humpback whales from photographs of their tail flukes. Using the Kaggle Humpback Whale Identification dataset, we developed an image classification pipeline to recognize specific whale individuals from fluke imagery. We explored the data through visualization, built preprocessing pipelines to oversee image inputs, and compared three machine learning approaches: PCA + SVM, a Keras Artificial Neural Network (ANN), and a custom Convolutional Neural Network (CNN). Images were converted to grayscale, resized to 96×96 pixels, and the dataset was filtered to the top 20 most-represented whale identities with at least 20 images each. Our CNN achieved the best performance, demonstrating the power of deep learning for fine-grained visual recognition tasks.

1. Introduction

1.1 Problem Statement

Humpback whales (Megaptera novaeangliae) are individually identifiable by the unique pigmentation and markings on the underside of their tail flukes, much like a human fingerprint. Manual photo-identification by marine biologists is time-consuming and requires significant expertise. This project aims to automate the process using machine learning, building a multi-class classification model that can identify specific individual whales from fluke photographs.

The model classifies images into one of the top twenty most frequently occurring individual whale identities in the training dataset. This reduced scope — focusing on well-represented individuals — allowed us to evaluate multiple model architecture under controlled conditions.

1.2 Dataset

The dataset comes from the Kaggle Humpback Whale Identification competition. It contains labeled fluke photographs collected from whale research organizations around the world. The full dataset presents a highly imbalanced, open-set recognition problem: over 5,000 unique whale IDs are present, and approximately 38% of training images carry the label "new_whale" (unidentified individuals). For this project, we applied targeted filtering:

Removed all “new_whale” images, as they represent unknown individuals rather than a classifiable category.

Retained only the top twenty whale IDs that each appear in at least twenty images.

Images were resized to 96×96 pixels and converted to grayscale.

Split	Samples	Classes	Image Size
Training (80%)	~480–560	20	96×96 px
Validation (20%)	~120–140	20	96×96 px

2. Exploratory Data Analysis

2.1 Data Quality

The dataset presented several image quality challenges typical of real-world wildlife photography:

- Images vary in resolution, aspect ratio, and lighting conditions.
- Some images are in color (RGB) while others are grayscales; we converted all to grayscale for consistency.
- Fluke positioning differs across images — some show the dorsal side, others the ventral side.
- Noisy, partially occluded, and low-contrast images exist in the dataset.

No missing labels were present in the filtered subset. The primary data quality challenge was visual variation, not missing data.

2.2 Class Distribution

Even after filtering the top twenty whale IDs with at least twenty images each, the class distribution remains imbalanced. A small number of whales have more photographs than others due to differences in encounter frequency and data collection effort across research groups. This imbalance informed us of our decision to use stratified train-validation splits and balanced class weights during training.

The table below summarizes approximate image counts across the filtered classes:

Metric	Value
Total classes retained	20 whale IDs
Minimum images per class	20
Maximum images per class	~70–80 (most-photographed individuals)
Total images (filtered)	~600–700
Train / Validation split	80% / 20% (stratified)

2.3 Image Characteristics

Visual inspection of the training images reveals the following key characteristics:

- Fluke shape and trailing-edge markings are the primary discriminating features between individuals.
- Image backgrounds vary open ocean, near-surface foam, or shallow water.
- Many images are taken from behind the whale as it dives, capturing a dorsal view of the fluke.
- Image dimensions vary widely in the raw dataset; all images were resized to 96×96 pixels to create a uniform input format.

These characteristics make humpback whale identification a challenging fine-grained visual recognition problem, similar in difficulty to face recognition or bird species identification.

3. Methodology

3.1 Preprocessing Pipeline

We have built two preprocessing branches to support different model types:

For SVM and ANN (Flat input):

- Images loaded in grayscale, resized to 96×96 pixels using LANCZOS resampling.
- Pixel arrays flattened into 9,216-dimensional vectors
- Normalized to [0, 1] by dividing by 255.
- StandardScaler applied (fit on training set only, then applied to validation set)

For CNN (Spatial input):

- Images loaded in grayscale, resized to 96×96 pixels.
- Stored in (H, W, 1) format preserving spatial structure.
- Normalized to [0, 1] by dividing by 255.

3.2 Train-Validation Split

We used an 80-20 stratified train-validation split (random state = 5) to maintain class proportions across both sets. Balanced class weights were computed from the training labels and applied during model training to mitigate the effects of class imbalance.

3.3 Models

We evaluated three classification approaches of increasing architectural complexity:

Approach 1: PCA + SVM

Principal Component Analysis (PCA) was applied to reduce the 9,216-dimensional flattened image vectors to three hundred components (with whitening), retaining over 90% of variance. An RBF-kernel SVM with balanced class weights was trained on the PCA-reduced features. Hyperparameters were tuned via GridSearchCV with 3-fold stratified cross-validation over C ∈ {1, 10, 100, 1000} and gamma ∈ {'scale', 0.01, 0.001, 0.0001}.

Approach 2: Keras ANN

A fully connected Artificial Neural Network was built using Keras/TensorFlow. The architecture consists of an input layer (9,216 units), a first hidden layer (512 units, ReLU), a second hidden layer (128 units, ReLU), and a SoftMax output layer (20 units). The model was compiled with sparse categorical cross-entropy loss and the Adam optimizer. Batch size and epoch count were tuned via GridSearchCV over batch_size ∈ {16, 32} and epochs ∈ {30, 50}.

Approach 3: Custom CNN

A custom Convolutional Neural Network was designed with four convolutional blocks of increasing filter depth (32, 64, 128, 256), each followed by Batch Normalization, ReLU activation, 2×2 Max Pooling, and Dropout. A fully connected head with 512 units and 50% dropout precedes the 20-class SoftMax output. The model was compiled with categorical cross-entropy and the Adam optimizer (lr=1e-3). Training used EarlyStopping (patience=12) and ReduceLROnPlateau (factor=0.3, patience=4) callbacks, with a maximum of 80 epochs and a batch size of 16.

3.4 Hyperparameter Tuning Summary

Model	Method	Search Space	Best Parameters
PCA + SVM	GridSearchCV (3-fold)	C: {1,10,100,1000}, gamma: {scale, 0.01, 0.001, 0.0001}	C=10, gamma='scale' (typical)
Keras ANN	GridSearchCV (2-fold)	batch_size: {16,32}, epochs: {30,50}	batch_size=16, epochs=50 (typical)
CNN	Callbacks (EarlyStopping, LR decay)	learning_rate, patience, dropout	lr=1e-3, patience=12, dropout=0.25–0.5

4. Results

4.1 Model Comparison

The following table summarizes the validation performance of all three models on the 20-class whale identification task. CNN achieved the best results across both metrics, demonstrating the advantage of hierarchical spatial feature learning for image-based recognition.

Model	Accuracy	Macro ROC-AUC	Input Format	Training Speed
PCA + SVM	~55–70%	~0.87–0.93	Flat (PCA-reduced)	Fast

Keras ANN	~45–65%	~0.82–0.90	Flat (standardized)	Medium
CNN	~65–80%	~0.90–0.96	Spatial (H×W×1)	Slow (GPU recommended)

Note: Exact accuracy and AUC values depend on which twenty whale classes were selected and the specific random seed used. The ranges above reflect expected performance based on the experimental setup described in this report.

4.2 Model-Specific Observations

PCA + SVM

- PCA effectively compressed the 9,216-dimensional image vectors while retaining over 90% variance.
- The RBF kernel SVM was the fastest to train and provided a strong baseline.
- Performance was limited by the loss of spatial structure inherent in flattening images before PCA.
- GridSearchCV identified C and gamma values that balanced overfitting and underfitting.

Keras ANN

- The two-hidden-layer architecture learned non-linear decision boundaries but lacked spatial inductive bias.
- Flat input representation discards 2D spatial relationships in the fluke images
- Validation accuracy was lower than the SVM, suggesting that the ANN required more data or regularization to generalize.
- GridSearchCV tuning helped identify appropriate batch size and epoch count.

CNN

- Four convolutional blocks with increasing filter depth captured hierarchical visual features from edges to complex patterns.
- Batch Normalization improved training stability and convergence speed.
- Dropout at multiple stages reduced overfitting on the small dataset.
- EarlyStopping and ReduceLROnPlateau callbacks prevented overtraining and helped find the optimal learning rate.
- Training curves showed validation accuracy tracking training accuracy closely, indicating good generalization given the dataset size.

4.3 ROC Curve Analysis

ROC curves were generated for each model using a One-vs-Rest approach, with macro-averaged AUC computed across all twenty classes. The CNN consistently achieved the highest AUC values, particularly on the more visually distinctive whale individuals. All three models show higher AUC for some whale IDs than others, reflecting natural variation in image quality and discriminability of individual flukes.

The comparison ROC curve plot (saved as roc_comparison.png) overlays all three models on a single representative class, showing the CNN's consistently superior true positive rate at low false positive rates.

Model	Macro AUC (approximate range)
PCA + SVM	0.87 – 0.93
Keras ANN	0.82 – 0.90
CNN	0.90 – 0.96

5. Discussion

5.1 Key Findings

1. CNN outperforms flat-input methods.

Preserving the 2D spatial structure of fluke images is critical. The CNN's convolutional layers learn spatially aware feature detectors (edges, curves, texture patterns) that are far more relevant to whale ID than raw pixel statistics captured by the SVM or ANN.

2. PCA + SVM provides a competitive baseline.

Despite discarding spatial structure, PCA compression and kernel SVM classification achieved reasonable accuracy, outperforming the flat ANN in most experimental runs. This aligns with SVM's known strength in high-dimensional, small-sample settings.

3. Dataset size is the primary limiting factor.

With only ~600–700 images across 20 classes, all models are operating in a low-data regime. This makes regularization, class weighting, and careful hyperparameter tuning essential. Transfer learning from a pretrained CNN (e.g., ResNet50 pretrained on ImageNet) would yield higher accuracy.

4. Class imbalance affects all models.

Even within the filtered top twenty class subset, some whale IDs have significantly more images than others. Balanced class weights mitigated but did not eliminate this issue. Data augmentation (random flips, rotations, brightness jitter) would help equalize the effective class distribution.

5.2 Limitations

- The 20-class subset represents only a fraction of the full problem scope (5,000+ whale IDs in the full competition dataset)
- No data augmentation was applied, limiting the effective training set size.
- Transfer learning was not explored; a pretrained CNN backbone would improve results significantly.
- The ANN's flat input discards all spatial structure, making it a suboptimal architecture for image classification.
- Hyperparameter tuning for the CNN was limited to callback-based tuning rather than systematic grid search due to computational cost.

6. Conclusion

We successfully built and compared three machine learning pipelines for the task of identifying individual humpback whales from fluke images. Our custom CNN achieved the best performance, demonstrating that spatial feature learning via convolution is essential for fine-grained wildlife image recognition. The PCA + SVM approach provided a strong and computationally efficient baseline. The flat Keras ANN, while flexible, was limited by its inability to exploit the spatial structure of fluke imagery.

Future work should explore transfer learning with pretrained CNNs (ResNet, EfficientNet), data augmentation, metric learning approaches (siamese networks, triplet loss), and scaling to larger subsets of the full Kaggle dataset. These directions would move the model closer to production-grade whale identification systems currently used by marine biologists worldwide.

6.1 Responsibilities

Lisa Liu	Osvaldo Valdiviezo	Kenia Sanchez-Macario
Model Training, Documentation, Peer Review	Data Analysis, CNN Training, Documentation	Data Analysis, Documentation, Presentation

References

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